

Let's consider the generic Markov Decision Process (MPD) $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$ and a generic policy π . Let's consider the Reinforcement Learning problem.

- Provide the pseudo-code for the Monte Carlo Policy Evaluation algorithm to estimate V_π in the generic MDP; in the algorithm consider
 - the 'first-visit' version;
 - the incremental estimation for quantities evaluation based on the step size α .
- Provide the pseudo-code for the TD(0) algorithm to estimate V_π in the generic MDP; in the algorithm consider
 - the incremental estimation for quantities evaluation based on the step size α .
- Monte Carlo and TD(0) are two estimators of different quantities: comment on the differences between the two estimators, their objectives and the advantages/disadvantages.



- Consider the MDP in the Figure above, the states reported in the Figure (considering a single starting state and two terminal ones): the action space is $\mathcal{A} = \{\text{left}, \text{right}\}$ and the state transitions are the obvious ones. The rewards associated to the couple (state, actions) are the ones reported in the Figure, that can be seen associated to the edges between state s_t and state s_{t+1} . Consider a random policy π , ie:

$$\pi(\text{left}, s_j) = \pi(\text{right}, s_j) = 0.5 \quad \forall s_j \in \mathcal{S}.$$

Consider the following two episodes generated by π :

- Episode #01 - (B, right); (C, right); END
- Episode #02 - (B, right); (C, left); (B, left); (A, left); END

Estimate V_π by using the Monte Carlo algorithm described at point 1 and considering:

- $\gamma = 1$;
- initial estimation $V_\pi(s) = 0.5$ for all non-terminal states;
- $\alpha = 0.5$

Report the procedure with calculations.

- Estimate V_π by using the TD(0) algorithm described at point 2 and considering the same settings at point 4. Report the procedure with calculations.
- With this peculiar choice of π it is quite easy to estimate the 'true' V_π even without applying Reinforcement Learning algorithms. Compare the estimations of V_π previously obtained with Monte Carlo and TD(0) with the 'true' V_π : which algorithm better estimated the 'true' V_π .
- Define, also with illustrations, the generic Generalized Policy Improvement approach.
- Suppose to apply Policy Improvement after the second episode and based on the estimations obtained at point 4 and 5: define a new policy π' .
- Suppose you want to evaluate a second policy a random policy μ , ie:

$$\mu(\text{left}, s_j) = 0.25, \quad \mu(\text{right}, s_j) = 0.75 \quad \forall s_j \in \mathcal{S}.$$

Can the episodes generated at point 4 with π be used to estimates V_μ ?

- (OPTIONAL) Estimates V_μ with the episodes generated at point 4 with π .

REPORT YOUR ANSWERS IN THE PROTOCOL SHEET BY CLEARLY NUMBERING THE RESPONSES TO THE VARIOUS POINTS